

GUISSTANN Project

Internship Report

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1. Introduction

1.1. Description of the project

GUISSTANN, started in 2023, is a joint project between the Center for Research in Economics and Statistics (CREST), Senegal's public Agence Nationale de Statistique et de la Démographie (ANSD), and the telecommunication private company Sonatel. The project aims to leverage mobile data to assist national statistics. In fact, it's aim is to predict census indicators, classifying them between:

- those that are easy to predict, and from which it would be relatively easy to build public policies
- those that require a more in-depth ground survey
- those that cannot be predicted by mobile data

The goal of GUISSTANN is to assist a developing country in resource allocation and in building fast humanitarian, financial and crisis responses by identifying and locating socio-economic situations.

It is heavily based on Blumenstock's 2015 paper [1], which describes a similar process in Rwanda.

1.2. Our mission

By the time I got to Dakar, the collaboration between the different institutions was well-established, but details still had to be discussed. In fact, even though we had access to as much mobile data as we wanted, we only had access to 10,000 household's pre-census data. It is enough to run a full pipeline, but not to produce results that will factually assist national statistics.

My mission was thus to leave Dakar with a fully functioning pipeline and ways to assess results. I also did a few meetings to consolidate institution relationships and train employees on potential techniques that could be used in their collaboration (split learning, white station).

2. Methodology

Below is the description of the pipeline produced, going from raw data to actual predictions, assuming all files are on a single physical computer.

2. 1. Data Description and Preprocessing

A. ANSD Data Preprocessing

The ANSD dataset comprises structured census data collected at the household level. To begin the preprocessing pipeline, we harmonize household identifiers across sections by standardizing the variables IDDR, ID_EDIF, and MEN_NUM. In order to get a complete and consistent list of values, we drop values for household duplicates, missing hashed phone numbers, as well as high-cardinality columns to prevent overfitting. Each section of the survey undergoes specific cleaning and standardization procedures. To a larger extent, binary columns are converted to boolean values, and categorical columns are encoded into integer category codes. This step outputs cleaned binary and categorical ANSD CSV datasets for each section, a JSON mapping file for categorical labels, and a consolidated list of valid hashed phone numbers.

B. Sonatel Data Preprocessing

The datasets provided by the Sonatel operator include call detail records (CDR), transactions, mobile data, and recharges. As a preliminary step, we apply a geolocation mapping with antenna coordinates to identify and standardize towers. Then for each data file, we retain only entries whose phone hashes appear in the valid ANSD list (A), timestamp them using date and time columns, and drop duplicates. For the CDR file, additional preprocessing steps include the conversion of binary columns to boolean values, filtering of unknown antennas, mapping of known antennas to tower IDs by callers and recipients (a step also done for mobile data), and the classification of call records to obtain full communication histories. This step outputs cleaned CDR, mobile data, transaction, and recharge files, along with a file with all encountered hashed phone numbers that match the valid list.

C. Combination of valid entries

To unify the census and mobile datasets we make a combined list of all hashed phone numbers from the ANSD and Sonatel files, and drop duplicates and missing values for unicity. We then assign to each entry a unique integer identifier in order, prioritizing those that exist in the ANSD dataset then adding the remaining ones from the Sonatel dataset. We output a combined mapping table containing all matched identifiers, as well as a filtered version that retains only the phone numbers

present in the ANSD dataset. This filtered mapping will be used for supervised learning tasks, where ground truth census indicators from the ANSD survey are necessary.

D. Harmonization of identifiers across datasets

Finally, we apply the mapping to all preprocessed ANSD and Sonatel files, thereby replacing each hashed phone number with its identifier. This harmonized system enables consistent referencing across datasets and facilitates downstream statistical modeling and analysis.

2. 2. Feature Extraction

Our objective with feature engineering is to convert an individual’s mobile phone metadata, from Sonatel dataset, into a set of quantitative measures that can be used to predict census data. In this phase, the inputs are the priorly cleaned files for each phone metadata section. These are loaded alongside the configuration file which is responsible for setting general parameters, definitions and mechanisms. As such, we set log level, blocks for different phone metadata sections, the access paths to relevant files, boundaries for cdr and speed, the number of phone numbers processed at once, and the standard column mappings.

Our next step consists of the enrichment of the data. At this point we apply temporal masks to all phone metadata sections (CDR, mobile data, mobile money) to differentiate between day and night time, and weekday and weekend. For the CDR and mobile money files, we also create a boolean column ‘initiated’ so that we can then reverse the dataset by swapping this column from True to False or vice versa. The original and reversed datasets are merged, hence including duplicates, but ensuring that communication is registered in both directions. Only for the CDR file, we also sort calls into voice and international, filter by caller id and timestamp, apply kalman smoothing and, lastly, create caller and recipient density features. In this manner, we obtain enriched and temporally structured datasets that capture user behavior more comprehensively and lay the foundation for robust feature extraction.

Finally, we perform feature extraction. For CDR, we create features for active days, event counts by period, soft cdr masks, duration and entropy features, international features, number of unique contacts per period, percent top 3 contacts, response delay rate text, entropy of contacts, balance of contacts, interactions per contact, intervenant time, interactions per contact, intervenant time, intervenant time daily, peak activity hour, distribution entropy, percent pareto durations, number of towers, entropy of towers, percent at home, percent at main day tower. As geographical features, we define distance smoothness, speed features, radius of gyration, location diversity, soft mobility masks, clustering features, rapid

sequence features, density features. For mobile data and recharges we calculate mobile data and recharges amounts respectively. For mobile money, we create stats for mobile money amount and mobile money contact, as well as mobile money transaction counts. The last stage of feature extraction involves merging all the features to create a table with all caller ids followed by the values of all the extracted features. This unified feature table serves as the final output of the pipeline, providing a comprehensive and standardized representation of user behavior ready.

2. 3. Modular Deep Learning Library

We use a deep learning framework to make our predictions. The inputs to this system are the preprocessed datasets (I) in the form of exploitable tabular data. X is all of the extracted features (II) and Y the target census indicators. For each indicator, we independently fit the following series of supervised learning models: multilayer perceptrons (MLPModel), gradient-boosted decision trees (XGBoostModel and LightGBMModel), random forests (RandomForestModel), L1-regularized logistic regression (LogisticLassoModel), support vector machines (SVMModel), and single-layer perceptrons (PerceptronModel). These models are implemented in a modular architecture, which allows for experimentation and consistent evaluation.

Thus each model is trained and evaluated with accuracy, R2 score, F1 score, area under the ROC curve (AUC), and confusion matrix metrics. This allows us to obtain the aggregated R2 score, computed as the average R2 across all runs for a given indicator weighted by the number of observations. It offers a standardized evaluation of model performance across different indicators, enabling better comparison between models despite variations in their predictive capacity for different target variables. From there, we determine which model performs best for each indicator. In addition, we extract feature importance scores to interpret the relative contribution of each input variable.

2. 4. Feature Selection Experiments

Our approach in this section builds on prior work, particularly by Blumenstock (2015) and Aiken et al. (2022, 2023). Both studies highlight the tradeoff between capturing variation and maintaining interpretability and intuitiveness, which is largely determined by the number and type of features used in machine learning models. Drawing on this insight, once we complete feature extraction and define the machine learning algorithms, we experiment with different subsets of features to evaluate their impact on model performance.

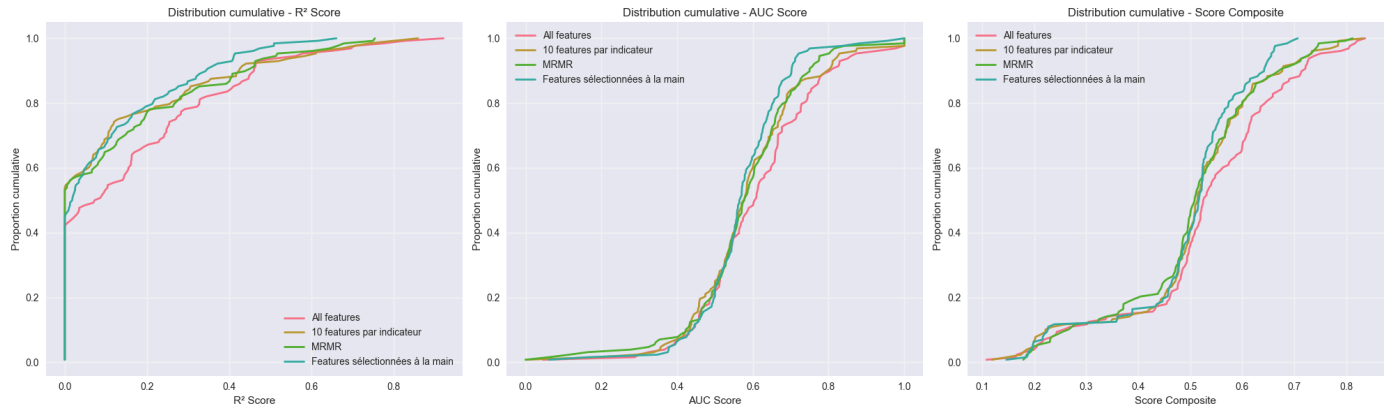
In our baseline model, we use all features with the intention of capturing as much variation as possible from the phone metadata. However, this implies less interpretability and longer computation and processing times. Addressing these issues, we experiment with three other subsets of features.

Each experiment involves re-running all supervised learning models for each indicator, but restricting the input space to only 10 features, and we evaluate them using the same modeling pipeline and performance metrics as the baseline, for direct comparison.

- The first experiment uses a set of 10 manually selected features identified as the most intuitive. These features are chosen by plausible links to socioeconomic indicators. For example, features related to total call volume, average daily mobility, night-time activity, and mobile money usage were prioritized. The goal here is to assess how well human-interpretable signals perform relative to more statistically optimized alternatives.
- In the second experiment, we select the 10 most relevant features based on their importance scores for each indicator, computed during the baseline modeling phase. This method is data-driven and aims to identify features that are individually strong predictors, regardless of redundancy or correlation with one another. It also enables us to verify the necessity of a large quantity of features.
- For the third experiment, we use the Minimum Redundancy Maximum Relevance (mRMR) criterion to select the 10 most informative global features. These features are chosen for their high correlation with the target variable (maximum relevance) while also ensuring low mutual information or correlation among themselves (minimum redundancy). After selecting these top 10 features using mRMR, we train all models using only this reduced feature set. This approach helps capture complementary aspects of the input space and may improve generalization by reducing overfitting to redundant signals.

3. Result Analysis

After running the pipeline on the provided 10,000 households and their equivalent mobile data, we obtain the following results for R2 score, AUC score and composite score.



We are interested in predicting indicators that “work well”, aka have a good score with these metrics. Running the pipeline allowed us to come up with a list of 20 of the best-predicted indicators.

Individual in-depth lookup on those with “bad” scores gave us explanations as to their results. From this we made lists of indicators that require a more in-depth ground survey, a list of districts where surveys weren’t conducted very effectively, as well as a list of indicators that simply did not have a large enough sample size.

4. Discussion

With our pipeline well in-place and documented, we allowed a modular usage for the researchers working on the project as well as future interns. The next goal for the project would be to extend the list of well predictable indicators, which is achievable through working on the lists we made during initial result analysis.

Another method we looked into was using multi task lasso, where by learning jointly we can improve prediction accuracy and variable selection.

Through meetings with Sonatel and ANSD executives and employees, we also realized that we would need a solution in the eventually where problems to share data would persist long term. For this we tried implementing split learning on a little scale, with only a sample of the data. It works, but has important limitations: there are way larger time and space complexities due to the heaviness of the computations, and we can only learn with differentiable models which are not always the best. Although split learning is an interesting approach, it is thus still to be worked on to perform well in our context.

5. Conclusion

Our team was able to produce a working pipeline on an extendable scale, and a guideline for future improvements. We were a bit discouraged by the non-matching progress speed of administrative procedures compared to technical and scientific progress. Nevertheless we wish the best for this project whose potential we truly believe in, and will continue to follow its course over the next few years.

6. References

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